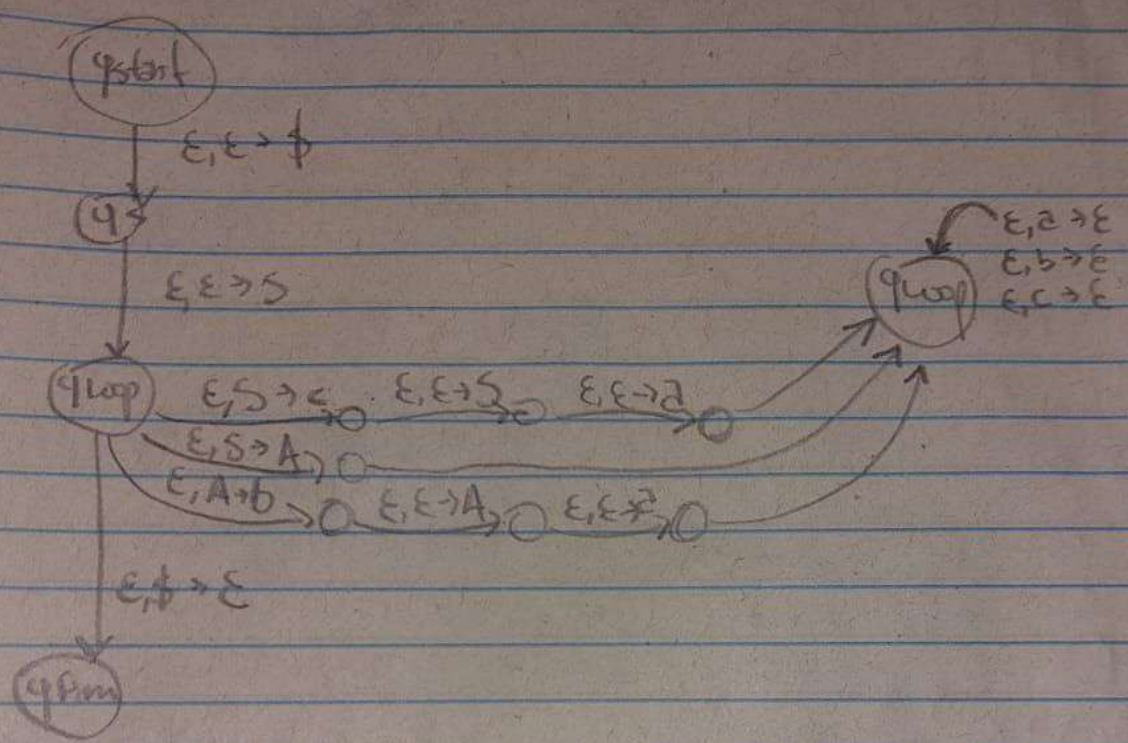


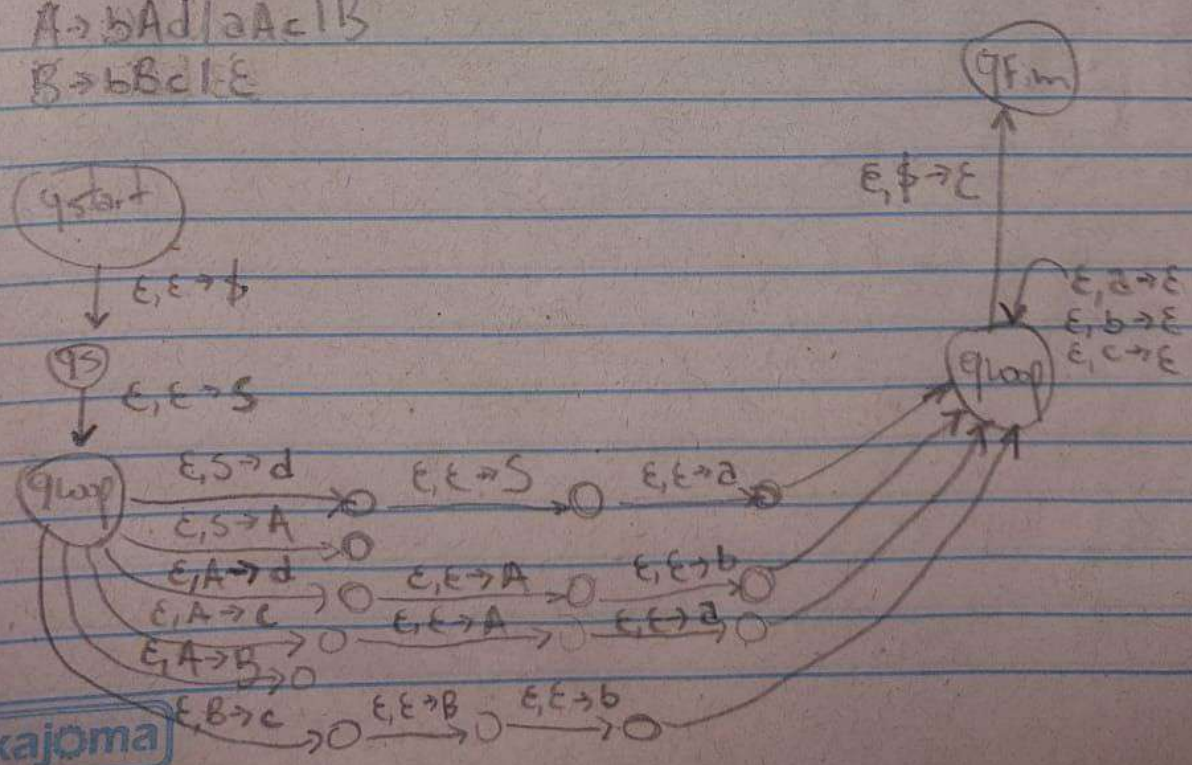
2. $L = \{a^i b^j c^k \mid i = j + k\}$

$S \rightarrow aSc \mid A$
 $A \rightarrow aAb \mid \epsilon$



3. $L = \{a^i b^j c^k d^h \mid i + j = h + k\}$

$S \rightarrow aSd \mid A$
 $A \rightarrow bAd \mid aAc \mid B$
 $B \rightarrow bBc \mid \epsilon$

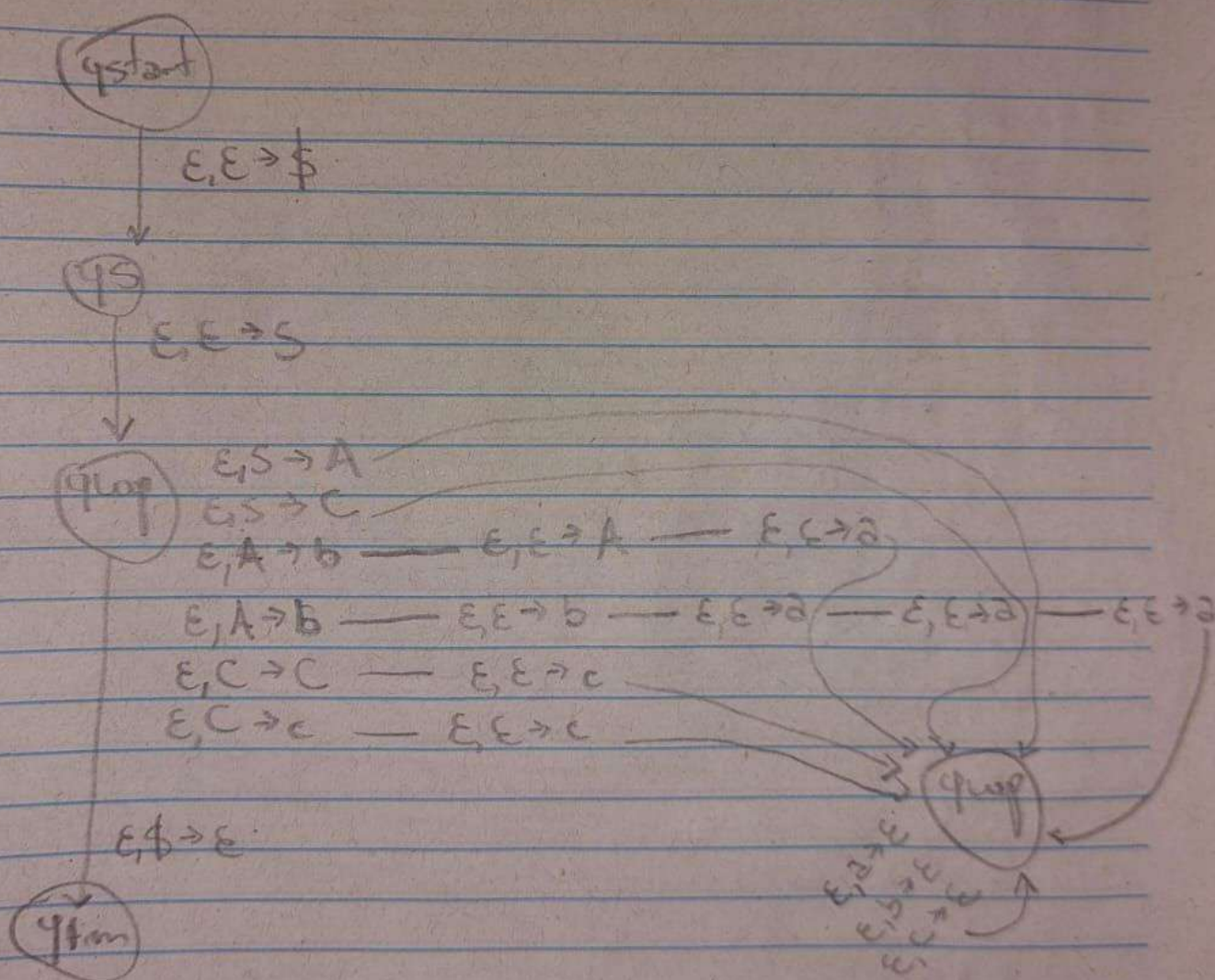


5. $L = \{a^i b^{\sigma} c^k \mid i \geq 3, \sigma = i-1, k \geq 2\}$

$S \rightarrow AC$

$A \rightarrow aAb \mid aaabbb$

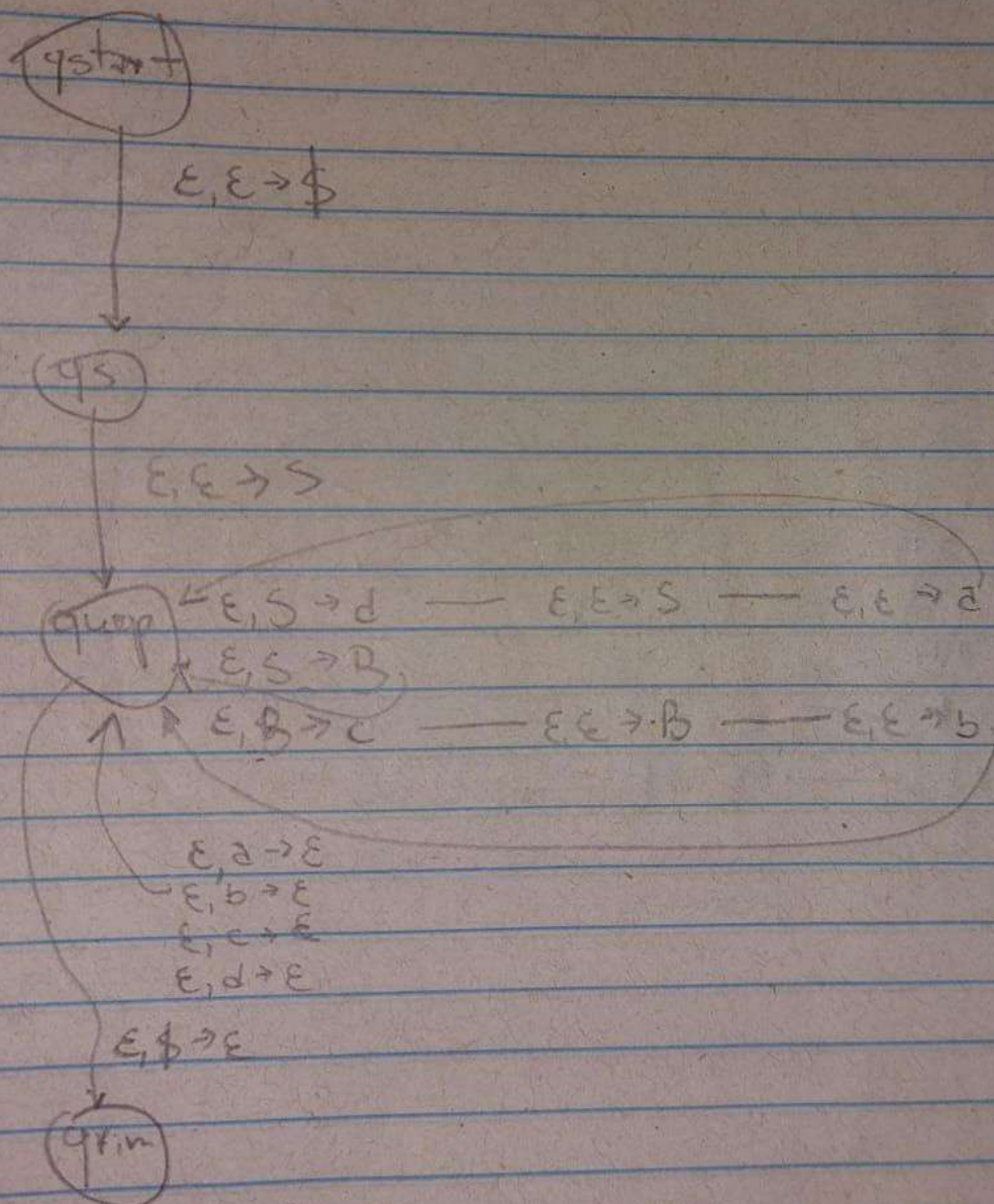
$C \rightarrow cC \mid cc$



6. $L = \{a^n b^m c^k d^l \mid n \geq 0, m \geq 0\}$

$S \rightarrow aSd \mid B \mid \epsilon$

$B \rightarrow bBc \mid \epsilon$

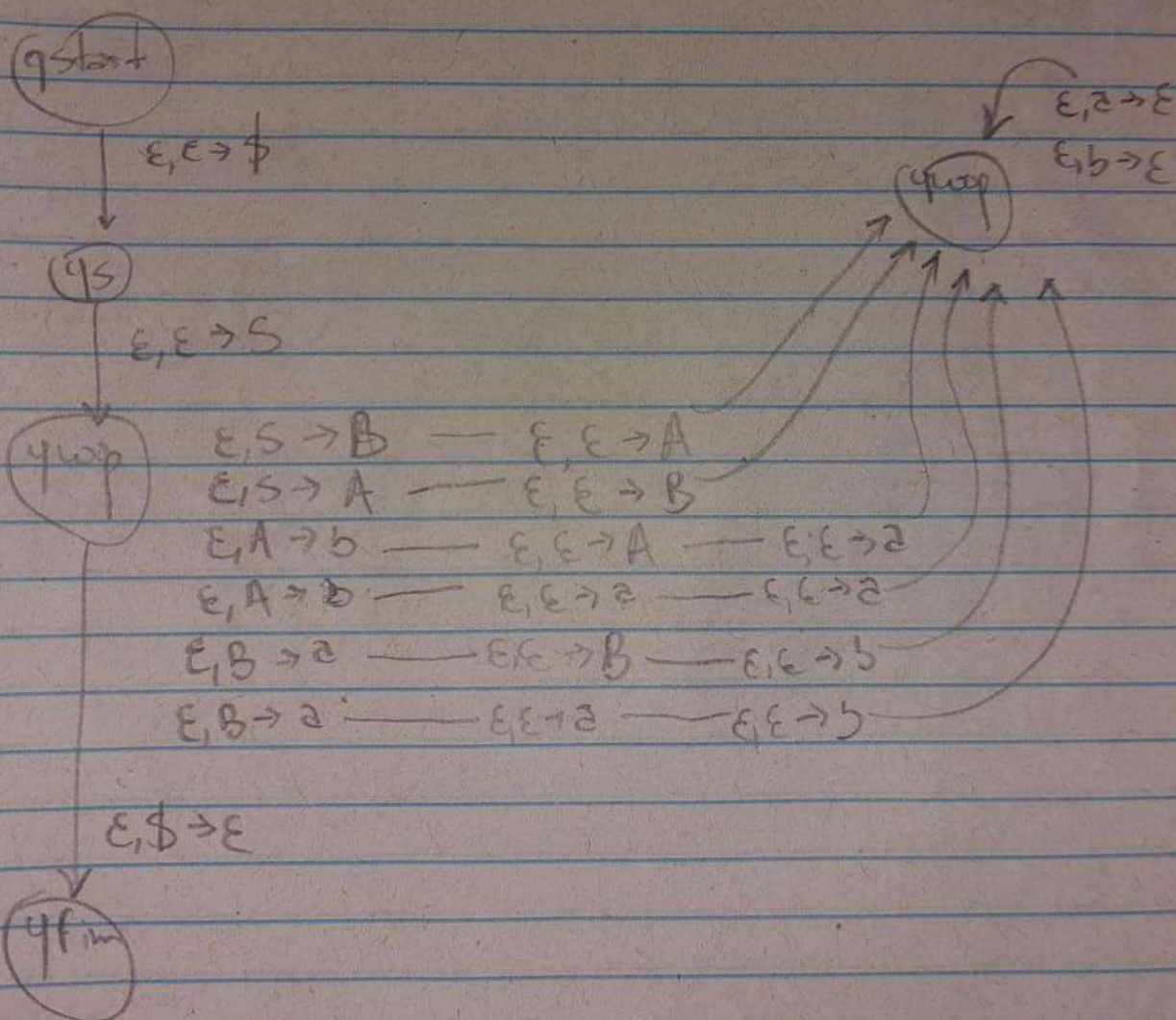


7. $L = \{ w \mid \text{contém um n.º par de "a" e ímpar de "b", independente da ordem} \}$

$S \rightarrow AB \mid BA$

$A \rightarrow aAb \mid aab$

$B \rightarrow bBa \mid baa$



aab
 aab
 aab

bba

aba

baa

baa

D S T Q Q S S

• •

8. $L = \{ w \mid \text{a diferença entre a qtd de } a \text{ e } b \text{ seja divisível por } 3 \}$

$S \rightarrow aP \mid bQ \mid \epsilon$

$P \rightarrow aQ \mid bS$

$Q \rightarrow aS \mid bP$

q_{start}

$\epsilon, \epsilon \rightarrow \epsilon$

q_s

$\epsilon, \epsilon \rightarrow S$

q_{loop}

$\epsilon, S \rightarrow P \quad \epsilon, \epsilon \rightarrow a$

$\epsilon, S \rightarrow Q \quad \epsilon, \epsilon \rightarrow b$

$\epsilon, P \rightarrow Q \quad \epsilon, \epsilon \rightarrow a$

$\epsilon, P \rightarrow S \quad \epsilon, \epsilon \rightarrow b$

$\epsilon, Q \rightarrow S \quad \epsilon, \epsilon \rightarrow a$

$\epsilon, Q \rightarrow P \quad \epsilon, \epsilon \rightarrow b$

$\epsilon, \epsilon \rightarrow \epsilon$

q_{fin}

q_{loop}

$\epsilon, a \rightarrow \epsilon$
 $\epsilon, b \rightarrow \epsilon$

9. $L = \{w \mid \text{que as subsequências de } a \text{ e } b \text{ é igual a de } ba\}$

$S \rightarrow aXa \mid bXb \mid a \mid b \mid \epsilon$

$X \rightarrow aX \mid bX \mid \epsilon$

